

Solving a new composition from a converged wind, and what the base handoff carries

1. What this memo records

Two things that are easy to confuse with each other.

Part 1 — the composition restart. How a converged EXHALE solution is used as the starting guess for a *different* composition: what the run script does, what `load_IC` does to the loaded densities, where the seed's information is kept and where it is overwritten, and — the point that matters for how results are quoted — why the seed sets only the path and never the answer. Everything in this part is implemented, verified against the source, and in production use in the LHS 1140 b composition and K_{zz} scans.

Part 2 — the base handoff. EXHALE now has two handoff routes: the scalar `base.inp` and the profile file named by `Lower atmosphere profile:.` This part records what each one is in the code today, the comparison on which the choice between them was made, and — item by item against the plan's requirement list — what the profile route actually carries, what reads each field, and what is still not transmitted.

Decision (2026-08-26), implemented 2026-08-27. Phase E goes to a profile file. The scalar `base.inp` is not retired: it stays as the single-layer route, and its behavior with no profile beside it is unchanged. What the decision left to the implementation — the schema, and what happens when both files are present — is now written: the schema is `docs/phase_e_flux_closure_design.md §2`, and a scalar physics key beside a profile is refused rather than merged (`src/modules/files_IO/input_read.f90:1371-1386`).

Part 1 — solving a new composition from a converged wind

2. The recipe, as the run script executes it

The production example is `LHS1140b/exhale/run_heh_diff_case.sh`, which solves one point of the helium-abundance scan with the binary H/He diffusion operator active. In order (line numbers are that file):

1. **Clone the input, change one line.** The control case's `input.inp` is copied with `sed` and exactly the composition line is replaced (lines 27–31): `He/H number ratio:` takes the new value, while the planet tag, the direct-steady switch and the residual target are set for this run and `He_Kzz:` is appended (line 32). Nothing else about the configuration moves — same spectrum, same domain, same flags — so the scan is a scan in composition and not in anything else.
2. **Hand the converged seed over as the initial condition** (lines 33–34): the seed run's `output/Hydro_ioniz.txt` and `output/Ion_species.txt` are copied into the new run directory as `output/Hydro_ioniz_IC.txt` and

output/Ion_species_IC.txt, which are the two file names load_IC reads (src/modules/files_IO/load_IC.f90:2-4, 89, 106, 126).

3. **Solve the steady state directly.** The binary is run with EXHALE_PTC=1, EXHALE_PTC_JFNK=1 and EXHALE_PTC_DTAU0=1.0 (line 43), i.e. the JFNK direct-steady route rather than a marching relaxation.
4. **Feed the output back as the next initial condition,** when the diffusion operator is on (lines 40–41): each further invocation copies the run's own last output over its *_IC.txt files. This is the outer Picard loop of composition and wind driven from the outside, one pass-set at a time, and the loop is stopped when the reported composition drift stops falling.
5. **Post-processing and transit** (lines 51–57): one Do only PP: True pass to write the advection-corrected profiles, then EXHALE_transit.py for the He I 10830 line.

Inside a single invocation the same Picard loop runs up to twenty times when He_diffusion is on and exactly once when it is off (src/EXHALE_main.f90:1238); it exits on a composition drift below 10^{-3} (line 1268) or on a failed steady solve (line 1254), and the composition update is under-relaxed with ω starting at 0.5 and halved down to a floor of 0.125 on any pass whose drift failed to fall (lines 1229, 1270–1274). The drift tested is the undamped distance to the fixed point, so a small ω cannot buy a false convergence (lines 1213–1214).

3. What the restart does to the composition

The seed file holds *absolute* species densities. Read as they stand, they would silently run the donor's composition while the setup report echoed the input one. The input file is the authority, so load_IC carries the loaded hydrogen and helium onto the requested ratio (src/modules/files_IO/load_IC.f90:176–192).

3.1 Counting the elements

In each cell the two element densities are counted over every species that carries a nucleus, molecular carriers included (load_IC.f90:193–199):

$$n_H = n(\text{H I}) + n(\text{H II}) + 2[n(\text{H}_2) + n(\text{H}_2^+)] + 3n(\text{H}_3^+) + n(\text{HeH}^+), \quad (1)$$

$$n_{\text{He}} = n(\text{He I}) + n(\text{He II}) + n(\text{He III}) + n(\text{HeH}^+). \quad (2)$$

HeH^+ is inside both counts, because it carries a nucleus of each. The metastable 2^3S state is *not* a term of its own: the HeI column the code transports and the file carries is the total He I population, the triplet included, so its nucleus is already counted there and adding it again would count it twice (load_IC.f90:196–197).

These are the integer weights bsp_nH and bsp_nHe of the species table (src/modules/init/species_table.f90:97–103: bsp_nH = 1, 1, 0, 0, 0, 0, 2, 2, 3, 1 and bsp_nHe = 0, 0, 1, 1, 1, 1, 0, 0, 0, 1 over the ten base species) taken together with bsp_is_excited_level, .true. for HeTR alone, which is what every budget sum skips (species_table.f90:116–118). The same count is what element_ratio_HeH applies in the code proper (src/modules/functions/composition.f90:211–239) and what the diffusion operator writes back against (src/modules/functions/binary_

element_diffusion.f90:776-799). One element count, one definition, three call sites — there is no second formula that can drift away from this one. That the triplet had to be excluded was measured, not assumed: it was being added a second time in all three places and in four more, and the mass, particle, nucleus and collision-partner sums were each wrong by that amount until 2026-08-27 (docs/Update_EXHALE.md §75).

3.2 The redistribution

Let HeH be the requested ratio. The deviation of the loaded state is measured cell by cell as $|n_{\text{He}}/n_{\text{H}} - \text{HeH}|/\text{HeH}$ and the redistribution is triggered when its maximum over the column exceeds 10^{-6} (load_IC.f90:200-207). A restart at the composition the file was written with therefore leaves every density untouched — bit for bit — which is what keeps the same-composition restart path exact.

When it does trigger, the two scale factors are (load_IC.f90:226-227)

$$s_{\text{H}} = \frac{n_{\text{H}} + n_{\text{He}}}{(1 + \text{HeH})n_{\text{H}}}, \quad s_{\text{He}} = \text{HeH} \frac{n_{\text{H}} + n_{\text{He}}}{(1 + \text{HeH})n_{\text{He}}}, \quad (3)$$

so that the new element densities in that cell are

$$n'_{\text{H}} = \frac{n_{\text{H}} + n_{\text{He}}}{1 + \text{HeH}}, \quad n'_{\text{He}} = \text{HeH} \frac{n_{\text{H}} + n_{\text{He}}}{1 + \text{HeH}}, \quad n'_{\text{H}} + n'_{\text{He}} = n_{\text{H}} + n_{\text{He}}. \quad (4)$$

That is the defining property of the operation: **the total nuclei count of each cell is conserved** and only the split between the two elements moves. Every hydrogen-bearing column is multiplied by s_{H} and every helium-bearing one by s_{He} (load_IC.f90:242-253), the trace metals following hydrogen because they are defined relative to a hydrogen nucleus, so

- the **ionization-stage split inside each element** is preserved exactly: $n(\text{H II})/n_{\text{H}}$, $n(\text{He II})/n_{\text{He}}$ and the 2^3S fraction all come through unchanged, since a single factor multiplies every stage of the element;
- the **velocity, pressure and temperature** profiles are the seed's, read straight off Hydro_ioniz_IC.txt (load_IC.f90:106-120);
- the **mass density is not** read from the file. It is rebuilt by calc_rho from the redistributed densities under the run's own mass policy, He 2^3S , molecular and (under eos_metals) metal mass included (load_IC.f90:402-422). It therefore does move with the composition, as it must: swapping hydrogen for helium at fixed nuclei count changes the mass. The seed supplies the kinematics and the thermal state; the mass follows the new composition.

3.3 Two refusals, and the diffusion-on exception

HeH^+ carries one nucleus of each element, so no single factor can set both counts. Two cases are therefore refused outright rather than approximated (load_IC.f90:207-225): a file carrying no helium at all (nothing to rescale), and a molecular file carrying HeH^+ whose composition disagrees with the input while He_diffusion is off — such a state must be restarted at its own composition.

With He_diffusion on the rule changes, because the cell-by-cell element split *is* the state being restarted: the diffusion operator produced it, and a separated profile is the physics, not a defect of the file. The rescale is then confined to the

base cells $1 - N_g \dots 1$, which is exactly where the operator itself holds a Dirichlet reservoir composition; every cell above keeps the helium fraction it was written with, $s_H = s_{He} = 1$ (load_IC.f90:228-241). In those base cells HeH^+ is projected onto the two element totals the same way `binary_element_diffusion` writes back: it takes the smaller of the two factors, $\min(s_H, s_{He})$, and whatever nuclei that leaves short of n'_H and n'_{He} are deposited into the neutral ground species $H I$ and $He I$ (load_IC.f90:254-278; the same projection at `binary_element_diffusion.f90:1516-1543`). The projection is non-negative by construction — the smaller factor can only reduce HeH^+ — and it conserves both element totals, at the cost of moving a little of the molecular ion into the neutral species.

Either way the run says on stdout which of the two it did, with the loaded top-cell ratio and the input ratio printed (load_IC.f90:279-289), so the restart is never silent.

3.4 Elements the file does not carry

Separately from the composition, an element whose stages the file does not carry — no column at all, or columns identically zero because the run that wrote them had metals off — is built from its abundance exactly as a cold start does, $n_X = ab_X n_H$ with everything in the neutral stage, and the substitution is reported (load_IC.f90:293-343). The column test alone would accept a metals-off file and then run with zero metal density while the base boundary condition still counted metals in the mass and particle budget, leaving an $O(1)$ residual in the first cells; the zero-sum test is what closes that.

3.5 Elements whose reservoir the handoff has moved

A third case sits between the two above and was added on 2026-08-27: the file carries the element, but the *reservoir* has moved since the file was written. That is exactly what a flux-closure iteration does — it changes the elemental ratios of the lower atmosphere from one iteration to the next — so the base boundary condition would use the new El/H while every cell above the base still held the old one, and the elemental budget $n_{El}/n_H = (El/H)$ could not close above the first cell (docs/Update_EXHALE.md §79-80).

The rule is a single factor on the whole column (load_IC.f90:345-400):

$$r_{El} = \frac{(El/H)_{handoff}}{(El/H)_{loaded, \text{ base cell}}}, \quad (5)$$

applied to every ionization stage of that element alike, so the loaded ionization split and the radial shape of the profile cancel out of it and only the normalization moves. El/H at the base cell is counted in nuclei — the element summed over its stages against the hydrogen nuclei of equation (1) — which is the convention of `element_ratio_HeH` and of `src/utls/element_budget.py`, so the factor is measured in the units the budget check reads it back in.

Which elements are touched is not a test on the numbers. An element is rescaled if and only if the lower-atmosphere handoff *states* its reservoir, and what records that is the door the value came in through: `set_element_abundance` is the only routine either handoff reader uses to set an abundance, and passing through it sets `memem_from_handoff(e)` (input_read.f90:1464-1487, flag allocated and cleared at input_read.f90:879-880). An abundance that reached the run from `metals.inp` alone never passes through that door, so it is not marked and its column is loaded untouched; a restart with no handoff at all leaves the loop doing nothing at all, which is what keeps every pre-existing restart bit

for bit what it was. Three further guards skip the element: one already rebuilt from the abundance by the block above (it *is* the reservoir already), a zero abundance, and a loaded column that sums to zero at the base cell.

Helium is deliberately outside this rule and keeps the older convention of sections 3.: the base cells are set to the input He/H and, with He_diffusion on, the column above keeps its diffused split.

Each rescaled element prints its own line — loaded ratio, stated ratio, factor, at enough digits that a factor of 1.0001 is legible, because that is the size of the step a converged closure iteration takes (load_IC.f90:393–399).

4. The seed is a guess; the answer is the new steady state

This is the distinction the whole recipe rests on, and it is worth stating plainly because the two are easy to conflate when a scan is quoted as a table of restarts.

The redistribution of section 3. is *arithmetic on a starting guess*. It is not a solution at the new composition, and it is not claimed to be one: it holds (v, p, T) and every ionization-stage ratio at the donor's values, which are the values that were in equilibrium with the *donor's* composition. What makes the result a result is what happens afterwards — the JFNK steady solve, the ionization-equilibrium solve, and (with diffusion on) the composition outer loop, iterated until the wind residual meets its target and the composition drift stops moving. The convergence *path* is what the seed choice buys; the converged state is a property of the new configuration alone.

The operational consequence: a restart chain is legitimate exactly as long as each rung is converged on its own terms, and each row of a scan must be quoted with its own achieved residual and drift, not with the seed's. That is why the LHS 1140 b tables carry info, $\|R\|$ and the composition drift in every row.

Two measured cases show that the seed choice is a real question of path and nothing more.

The helium scan needed no ladder. In the $K_{zz} = 10^9$ composition scan, every case restarted from the converged *diffusion-off* solution of the same composition, or from the neighboring composition already solved with diffusion on, and every case reached info = 0 in its first pass-set and then stopped moving (LHS1140b/kzz_decision.md, section 6). The recorded reason is physical: a diffusion-off solution is well mixed near the base, which is where the adopted K_{zz} also mixes, so the step the line search is asked to take is small. The same held at 10^8 and 10^{10} (*ibid.*, section 6.1).

The eddy coefficient did need one. Restarting a case at $K_{zz} = 10^8$ or above directly from the $K_{zz} = 0$ control asks the solver to absorb the whole helium column in one step: the line search stalls and the composition outer loop stops on its solver-failure guard after a single pass (*ibid.*, section 3). Rebuilding one decade at a time — ladder_kzz.sh restarts each case from the case one decade below it — fixes the seeding. It does not fix everything: on this wind JFNK settles at $\|R\|$ between 2.7×10^{-3} and 4.7×10^{-3} rather than the 1.0×10^{-3} the control reaches, so the upper rungs state a looser Resid tol and quote the achieved $\|R\|$ in the table. A target of 2×10^{-2} was tried and rejected, because there the solver stops responding to the eddy term altogether.

Same code, same operation, opposite verdicts on whether a ladder is required — which is the concrete demonstration that seeding is about the step the nonlinear solver is asked to take, and that no property of the converged answer follows from it.

5. Evidence that the restart is faithful

The diffused profile survives a round trip. Measured against the stored `examples/14_diffusion` output under `EXHALE_DUMP_IC=1`, which writes the state as loaded, before the first ionization-equilibrium solve: the loaded He/H profile reproduces the file's to 3.3×10^{-16} relative, worst cell, over all 504 cells (`docs/Update_EXHALE.md` §67). That is round-off, i.e. the `He_diffusion` branch of section 3. leaves the column it is meant to leave alone.

The defect it replaced. The same file through the pre-change binary comes back *flattened* to the input ratio: at $r = 1.05 R_p$ the file holds 1.616×10^{-2} and the load produced 8.333×10^{-2} , a factor 5.2 (*ibid.*). The old rule rescaled every cell to the global He/H whenever the file's composition differed from it, on the ground that the input file is the authority — correct with diffusion off, and destructive of the answer with diffusion on. Both numbers are quoted from that record and were not re-measured for this memo.

And the composition authority itself was a measured need. A He/H = 10 restart seeded from a He/H = 1 solution was measured to converge back to He/H = 1 everywhere before the redistribution existed (`load_IC.f90:180-182`) — the donor's composition ran while the setup report echoed the input one.

The diffusion-off path is unmoved. `make check` reported byte-identical output on every case of the matrix as it stood on 2026-08-25, which is the evidence that confining the rescale to the base cells changed nothing outside the `He_diffusion` branch (`docs/Update_EXHALE.md` §67, quoted; the matrix has since grown from five cases to seven, and was not re-run for this memo).

Part 2 — the base handoff: scalars and profiles

6. The scalar handoff

`read_base_inp` reads an optional `base.inp` from the run directory (`src/modules/files_IO/input_read.f90:1203-1388`). It recognizes six named keys and one family of element keys, and every key belongs to one of the categories the routine's own header lists, because the category is what says what the value is allowed to do to the wind:

key	unit	category	what it sets
T_base	K	EOS boundary	T ₀ , the base temperature
r_base	R_J	EOS boundary	R ₀ , the base radius
p_base	bar	EOS boundary	p_base_bar, the level all of the above refer to (default 10 ⁻⁶)
q_H2_base	ratio	EOS boundary	q_h2_base, the H ₂ volume mixing ratio that sets the base particle count through comp_ntot_bc, in place of the chemical-equilibrium fit. An EOS anchor, not a composition pin: nothing holds H ₂ there and no H ₂ profile is seeded from it
HeH_base	ratio	elemental reservoir	HeH; a positive value also sets thereis_He
<El>_H_base	ratio	elemental reservoir	X _{El} , the El/H nuclei ratio, through set_element_abundance, for any of the ten elements of the species table (e.g. O_H_base 4.90e-4). Overrides metals.inp for that element, and switches the metal system on if it is the only nonzero abundance
Kzz_base	cm ² s ⁻¹	boundary constraint	he_kzz, the eddy coefficient the element-diffusion operator imposes at the base; inert with He_diffusion off

Comment lines and unknown keys are ignored, with one comment parsed: # solution_id <hash>, which is what the pair check of section 7. reads. The call sits in input_read after run_lower_atm_prestep and *before* the derived constants, so the overrides take effect on everything computed from them, and an absent file is a no-op (input_read.f90:858-882). Two consistency messages are printed and not enforced: that q_H2_base and HeH_base must come from the same lower-atmosphere solution (input_read.f90:1325-1336), and that a p_base away from 1 μ bar means the base composition and the equilibrium H₂ fit both refer to that other level (input_read.f90:1337-1342). Those two stay messages deliberately: an internally inconsistent scalar handoff is the user's to judge. What is *not* left to a message is the pair of files — see section 7..

Kzz_base deserves a note, because the scalar it writes has changed underneath it. he_kzz now defaults to 0, not 10⁹: an eddy term is a property of the atmosphere being modelled, so it is stated rather than inherited (src/modules/init/parameters.f90:146-156). And he_kzz is no longer what the diffusion operator reads. Every site in binary_element_diffusion reads the array kzz_cell instead (parameters.f90:157-164; the three sites are binary_element_diffusion.f90:679, 1380, 1597), filled once the radial grid exists by eddy_diffusion_on_grid:

from the interpolated Kzz column of a profile when there is one, and from `he_kzz` in every cell otherwise (`src/modules/files_IO/lower_atmosphere_profile.f90:465-529`). The uniform case is exactly the old scalar — for a uniform array the face average $\frac{1}{2}(a+a)$ is a in IEEE double — so a run with no profile sees the number it always saw.

A real file, the one the `mol_base_handoff` regression case pins (`backup/regression/mol_base_handoff/base.inp`, comments elided):

```
T_base      1140.0
r_base      0.49
HeH_base    0.0793
Kzz_base    1.000e+09
q_H2_base   0.75
p_base      1.000e-05
```

Four of the seven default regression cases carry such a file — `mol_base_handoff`, `mol_metals`, `mol_lyman_werner` and `mol_diffusion` (`backup/regression/run_check.sh:57`) — so the scalar contract is pinned by the golden comparison and cannot be changed silently. Those four goldens were re-snapshotted on 2026-08-27, but for a reason that has nothing to do with the handoff: a molecular-network audit found the three-body term using a mass density where a number density belongs (`docs/molecular_chemistry_audit_he_rich.md`, `docs/Update_EXHALE.md §82`). The scalar contract itself is unchanged.

7. The profile handoff

The second route is the file named by `Lower atmosphere profile`: in `input.inp` (`input_read.f90:447-455`; user manual §3.8). It is a table over an interval of pressure rather than a single level, and the reader is `src/modules/files_IO/lower_atmosphere_profile.f90`.

What the file is. A header of `# key value` lines and then one row per level, deep to shallow. The header states the fingerprint (`solution_id`, `source_code`, `source_version`, `mechanism`, `stellar_flux`, `notes`), the three pressures (`p_match_bar`, `p_top_bar`, `p_deep_bar`), the closure iteration (`iteration`, `trial_flux_H`, `trial_flux_He`, `reached_steady_state`) and the column names (`lower_atmosphere_profile.f90:185-198`). Nine columns are required — `p`, `r`, `T`, `n_tot`, `rho`, `Kzz`, `q_H2`, `q_H`, `X_He` — and the element columns `X_<El>` are found by name, so the producers may order them as they like (`lower_atmosphere_profile.f90:328-340`).

What is refused. The reader refuses rather than warns on a non-positive or non-monotonic pressure, a table whose top does not reach above the matching level (`p_top_bar` $\geq$ `p_match_bar`, i.e. no overlap at all), a matching level outside the pressures the table covers, and a missing `solution_id` (`lower_atmosphere_profile.f90:344-382`). Interpolation is linear in $\log p$ and is never extrapolated; a matching level that falls on a node returns that node bit for bit, which is what lets a single-level profile reproduce the scalar handoff exactly (`lower_atmosphere_profile.f90:396-434`).

What EXHALE takes from it. Exactly five kinds of quantity, all through the same doors the scalar keys use (apply_lower_atmosphere_profile, input_read.f90:1392-1460): T_0 and R_0 from the T and r columns at the match, $p_{\text{base_bar}}$ from $p_{\text{match_bar}}$, $q_{\text{H}_2\text{_base}}$ from the q_{H_2} column, HeH from X_{He} , and each metal abundance from its $X_{\langle\text{El}\rangle}$ column through set_element_abundance — the same routine $\langle\text{El}\rangle_{\text{H_base}}$ goes through, which is why an element stated by either route is marked `melem_from_handoff` and picked up by the restart rescale of section 3.5. K_{zz} is taken as a profile, interpolated onto the radial grid into `kzz_cell`. The n_{tot} and ρ columns are required of the file but are *not* imposed: the base density is n_0 of `input.inp`, and the base particle count and mass follow from T_0 , He/H , q_{H_2} and the elemental ratios exactly as on the scalar path.

The two files together are enforced, not printed. With a profile in use, every `base.inp` key in the three physics categories of section 6. — EOS boundary, elemental reservoir, boundary constraint — is refused with error `stop`, because the profile already states that quantity at the matching level and a scalar accepted beside it would be a second source of the same number (`refuse_scalar_key`, input_read.f90:1371-1386). What a `base.inp` beside a profile may still carry is provenance, and there the pair is checked: a missing `# solution_id` line, or one that differs from the profile's, stops the run (input_read.f90:1344-1367). `He_Kzz`: in `input.inp` is a different file with a different contract and is not refused, but it is inert beside a profile and the run says so (input_read.f90:1449-1458).

What pins it. The `lower_profile` regression case, one of the seven defaults (`backup/regression/lower_profile/`): HD 209458 b with `He_diffusion: True`, the base state and q_{H_2} taken from the table at $p_{\text{match}} = 10^{-6}$ bar, the elemental reservoirs carried by the profile with `no metals.inp` in the directory — so the profile is the only source of the metal abundances — a non-constant K_{zz} column interpolated onto the grid in place of the inert scalar, and a provenance-only `base.inp` whose `solution_id` matches, which pins the accepting branch of the pair check. The worked example is `examples/17_lower_profile/`.

8. Scalar file against profile file

This is the comparison the choice was made on, brought up to what each route is in the code today.

	scalar base.inp	profile file
what is transmitted	six named numbers plus one El/H per element, all describing one layer	columns on the lower model's own pressure grid over an interval that reaches above the match
relation between the models	a handoff: the lower model's state at one level becomes the upper model's boundary	a matching over an interval both models cover, refused outright when the file does not reach above the match
flux continuity	not checkable. One layer's values carry no gradient, so nothing in the file constrains an elemental flux, and HeH_base stays a prescribed input while "the wind returns H:He" is an assumption dressed as a result	checkable, and measured. The elemental face flux $4\pi r^2(\rho X v + J)$ is reduced to a median and a relative spread over two radial windows and written to EXHALE_resolved.out whenever a profile is in use (binary_element_diffusion.f90:1784-1872); the criterion is the same radial invariance as T8 (docs/Update_EXHALE.md §69)
eddy coefficient	one constant, Kzz_base $\rightarrow$ he_kzz	the Kzz column, interpolated onto every cell as kzz_cell
molecular composition	one scalar, q_H2_base	q_H2 and q_H at their own levels, plus the diagnostic molecules; the ions of the EXHALE network are still not transmitted (section 9.3)
what it costs to produce	a keyword scan, unknown keys ignored	a schema with units and a grid convention, interpolation onto the EXHALE grid, a fingerprint and a stated failure policy — all written (src/utls/lower_profile_schema.py and the two adapters beside it)
failure mode	a silent mismatch: the two consistency checks are messages, not refusals (input_read.f90:1325-1342), so a handoff assembled from two different lower-atmosphere solutions runs without complaint — unless a profile is present, in which case the pair is checked by fingerprint	a stated one: a missing or unmatched solution_id, a scalar physics key beside the profile, a table that does not overlap the match — each an error stop with the reason printed

Three rows carried the decision.

Flux continuity is the whole point of Phase E. A steady lower-atmosphere chemistry solution cannot determine the hydrogen supply independently of its upper-boundary flux. Without a flux-continuity condition at a matching surface, HeH_base remains prescribed and the composition is an input wearing the clothes of an output (docs/lhs1140b_lower_atmosphere_plan_new.md §2). A scalar file cannot express the condition, because a single level has no gradient

and carries no flux. This is not a convenience argument; it is the difference between a coupling and an assumption.

A constant K_{zz} is the wrong shape. The judgment is already on record: “a *constant* K_{zz} is the wrong shape of assumption for this problem regardless of its magnitude: the molecular coefficient it competes against rises by nearly five decades between the base and $1.2R_p$, so no single constant can both mix the base and follow the molecular coefficient outward, while Lindzen’s own result is that the eddy coefficient should *turn over* rather than keep growing” (docs/eddy_diffusion_kzz.tex, summary). K_{zz_base} can transmit only a constant. The homopause — the one consequential thing K_{zz} does — is set where $K_{zz} = D$, so it is a property of the two profiles and not of either number alone.

The molecular scalar has the same defect one level down. $q_{H_2_base}$ is a single H_2 mixing ratio standing in for a composition that the molecular network resolves into H_2 , H_2^+ , H_3^+ and HeH^+ . It is enough to select a branch of the base particle count; it is not enough to hand over a photochemical solution. This is the one row of the three the profile route did not close: it transmits q_{H_2} and q_H at their own levels, but none of the three molecular ions (section 9.3).

9. The decision, and what was built on it

9.1 Decided 2026-08-26 (user decision)

1. **Phase E goes to a profile file.** The interface decision the plan left open (“whether this stays a scalar file or becomes a profile file is an interface decision to confirm with the user before implementation”, docs/lhs1140b_lower_atmosphere_plan_new.md Phase E part 1) is settled in favor of profiles, on the flux-continuity row of section 8. above all. The plan itself now records the phase as done (milestones E1–E5, 2026-08-27) with its three acceptance criteria and the numbers behind them.
2. **The scalar base.inp is kept**, as the single-layer route rather than a deprecated one. Three reasons, in order: four regression cases pin it and their goldens are the evidence that the molecular base is unmoved; a single-level handoff remains the right description when the lower model genuinely produces one (the analytic column of Tier 1); and the code path is small and already validated.
3. **The profile route is opt-in through a new key** (`Lower atmosphere profile:`), and its absence leaves the previous behavior exactly as it was. This is the standing house rule — a new physics option defaults to off so old goldens still reproduce.

9.2 What was implemented, 2026-08-27

All three hold, with one qualification on the third. The route is opt-in through `Lower atmosphere profile:` and an absent key is a no-op, so the diffusion-off and scalar-handoff paths are untouched by any of it. The qualification is that the four molecular goldens are *not* the ones they were at the time of the decision: they were re-snapshotted at the end of the same day for the three-body density fix of docs/Update_EXHALE.md §82, which is a different change with a different reason. The Phase E work did not move them.

What the decision left open, and where it now lives:

piece	where it is
the schema	docs/phase_e_flux_closure_design.md §2; docs/input_schema.md §2d; user manual §3.8
the reader	src/modules/files_IO/lower_atmosphere_profile.f90
the run key	Lower atmosphere profile: (input_read.f90:447-455)
the producers	src/utls/lower_profile_schema.py (format, elemental accounting, fingerprint) with photochem_to_lower_profile.py and vulcan_to_lower_profile.py behind one command line
the closure loop	src/utls/element_flux_closure.py, driven from the fluxes EXHALE measures and writes to EXHALE_resolved.out
the regression	backup/regression/lower_profile/, one of the seven defaults
the example	examples/17_lower_profile/

9.3 What the profile file carries, against what the plan asked for

The plan's list (*ibid.*, Phase E part 1) against the schema, field by field. "read by" names the code that consumes it; a field the file carries and nothing consumes is stated as such rather than counted as met.

asked for	field	read by
matching pressure and radius	header <code>p_match_bar</code> ; columns <code>p</code> , <code>r</code>	<code>p_base_bar</code> directly; <code>R0</code> from <code>r</code> at the match (<code>input_read.f90:1418–1424</code>)
temperature and total density	columns <code>T</code> , <code>n_tot</code> , <code>rho</code>	<code>T0</code> from <code>T</code> . <code>n_tot</code> and <code>rho</code> are required of the file but not imposed: the base density stays <code>n0</code> of <code>input.inp</code> and the base particle count and mass are built from <code>T0</code> , <code>He/H</code> , <code>q_{H2}</code> and the elemental ratios, as on the scalar path
elemental H and He abundances	columns <code>X_He</code> , <code>X_<El></code>	<code>HeH</code> and, through <code>set_element_abundance</code> , every <code>melem_ab</code> (<code>input_read.f90:1430–1443</code>)
the upward elemental fluxes	header <code>trial_flux_H</code> , <code>trial_flux_He</code> ; columns <code>F_H</code> , <code>F_He</code>	the header pair by <code>element_flux_closure.py</code> , through <code>EXHALE_resolved.out</code> . The <code>F_H/F_He</code> <i>columns</i> are written by the adapters and read by no Fortran routine: <code>EXHALE</code> measures its own elemental fluxes instead and reports them, which is the half of the loop it owns
molecular/atomic H partition	columns <code>q_H2</code> , <code>q_H</code>	<code>q_H2</code> → <code>q_h2_base</code> , the base particle count. <code>q_H</code> is required of the file and read by nothing
K_{zz} at the match	column <code>Kzz</code>	not the match alone — the whole column, onto <code>kzz_cell</code> (<code>eddy_diffusion_on_grid</code>)
selected molecular abundances the EXHALE network needs	<code>q_H2</code> , plus the diagnostic set (<code>q_H2O</code> , <code>q_CO</code> , <code>q_CH4</code> , ...)	<i>not met</i> . The <code>EXHALE</code> network resolves <code>H2</code> , <code>H2⁺</code> , <code>H3⁺</code> and <code>HeH⁺</code> ; the file carries none of the three ions and no molecular profile is seeded from it. The diagnostic columns are metadata — nothing consumes them — and the schema says so
source fingerprint	header <code>solution_id</code> , <code>source_code</code> , <code>source_version</code> , <code>mechanism</code> , <code>stellar_flux</code> , <code>iteration</code> , <code>notes</code>	the reader, and <code>lap_report_provenance</code> into <code>EXHALE_resolved.out</code>

The two further requirements the plan states are both met:

- **the same-solution consistency of the handoff pair is enforced**, not printed. With a profile in use a `base.inp` without a `# solution_id` line, or with one that differs from the profile's, stops the run, and a scalar physics key beside the profile stops it earlier still (`input_read.f90:1344–1367`, `1371–1386`). Note the scope: this is enforcement of the *pair*. A `base.inp` on its own is still checked only by the two printed messages of section 6.;
- **the transit tool reads the resolved configuration**. `read_input_parameters` prefers `EXHALE_resolved.out` next to `input.inp` and takes the planet radius, mass, equilibrium temperature, orbital distance, stellar mass

and He/H from it, refusing to fall back silently if the file is present but unreadable (`exhale_transit_lib.py`: 336–364). `EXHALE_resolved.out` is written after both handoff readers have run (`src/modules/files_IO/write_setup_report.f90`: 428–543), so a handoff that moves the base radius now moves the transit geometry with it — Phase B of the plan.

9.4 What is still open

- The molecular ions of the network are not part of the handoff, as above. A photochemical $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ solution at the base is still something EXHALE re-derives rather than inherits.
- The overlap window the design defines the closure on was measured on LHS 1140 b and rejected there: it is $\sim 0.003 R_p$ thick and the flux across it spreads by a factor of 13–35, so the driver falls back on the steady-flux window $r \geq r_{\text{esc}}$ and records that it did (`src/utls/element_flux_closure.py`, header; `docs/phase_e_flux_closure_design.md` §3.4). The fallback is reported rather than silent, but the window the design was written around is not the window the number comes from.
- `q_H2_base` and the profile's `q_H2` are the composition of the inflowing gas in both routes, no longer EOS anchors alone (§117 of `docs/Update_EXHALE.tex`, 2026-09-01). They set the base particle count *and* are imposed on the H_2 partition of the lower ghost species, the particle count then coming from that same species state. Before that change the two disagreed by 6.4% and the base ran 73 K above the T_0 the boundary condition claimed to impose.